IBM Z Datathon: Theme 3 - Wildcard

PROJECT DISPATCH

A solution to New York EMS response times.



THE PROBLEM

13 minutes

AVERAGE URGENT CARE AMBULANCE RESPONSE TIME IN NEW YORK

Legacy infrastructure cannot handle the low staffing and high call volume especially in underserved boroughs.

"PEOPLE ARE DYING BECAUSE THERE ARE NO STAFF,"

Anthony Almojera, a lieutenant paramedic recognising the ever increasing callout time.

"CLIMATE PROJECTIONS FOR NEW YORK CITY SHOWS SOME OF THE STATE'S LARGEST INCREASE IN EXTREME HEAT,"

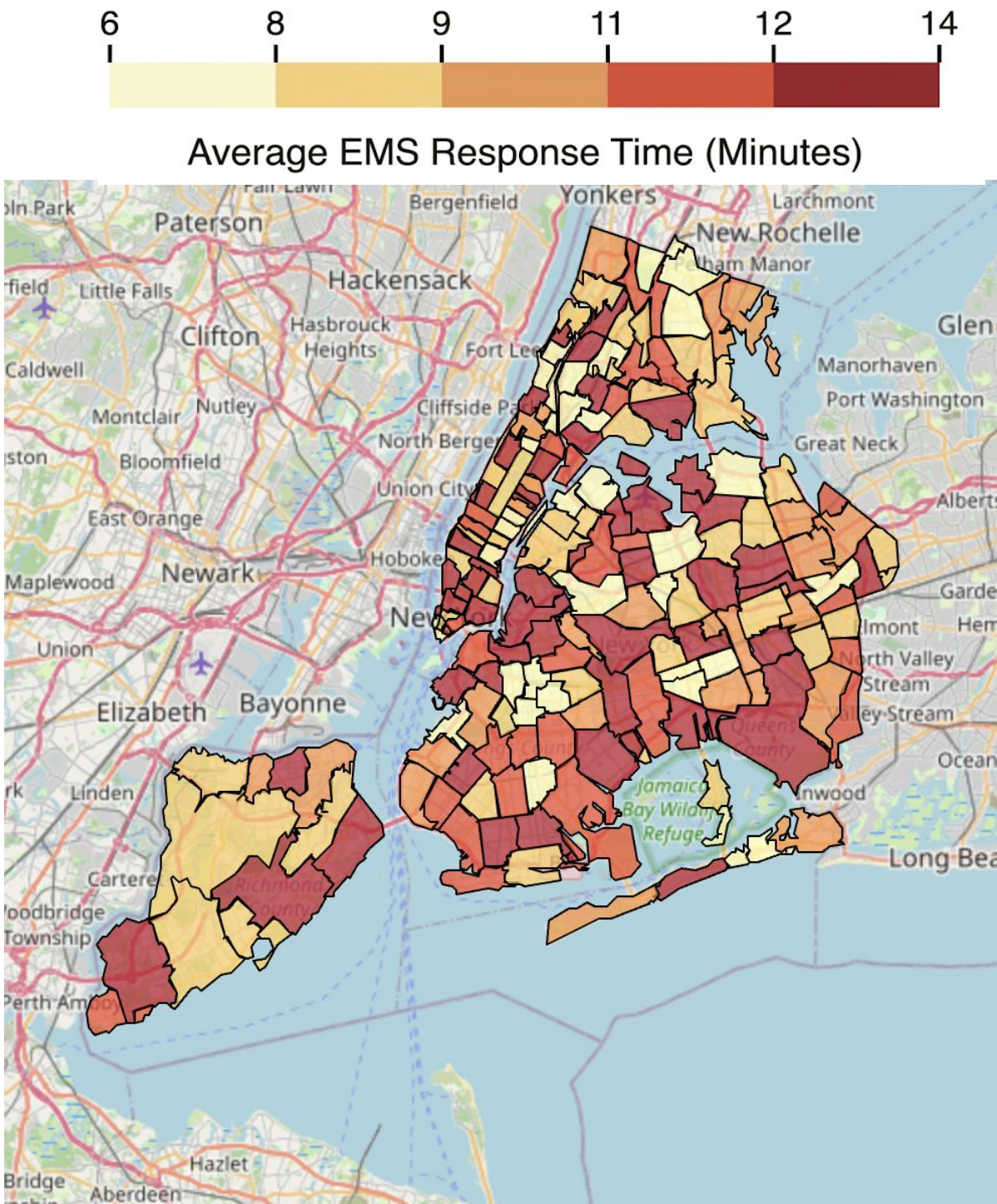
Climate change is creating more drastic weather, which effects the most vulnerable people.

THE TARGET DEMOGRAPHIC

PRIMARY ADOPTION FOCUSED ON DENSE METROPOLITAN REGIONS. MODEL TRAINED ON NEW YORK CITY AS IT HAS THE LARGEST AND MOST INFORMATIVE DATASET AVAILABLE.

490,000
VEHICLES

8.8M
NYC POPULATION



THE SOLUTION



MOBILE MEDICAL UNITS

Dynamic route optimization across NYC. The optimised number of units has been calculated to 330.



PREDICTIVE ROUTING

Historical incident-based unit distribution.



INCIDENT PREDICTION

Weather-triggered preventative medical deployment.



PRIORITY LOCATIONS

EMS dispatch data hotspot mapping.

THE IMPLEMENTATION

Our target client base is an emergency medical dispatch organisation, therefore we believe it is necessary to include all the statistics.

01

DATA

Real NYC 911 medical calls, plus weather, Census and hospital data, joined by ZIP zone and half-hour.

1.5M

911 calls from 2021 to train and validate the model

02

FORECAST

LightGBM predicts critical and other calls in every ZIP zone for the next 30 minutes to 3 hours.

8 models

tested on all of 2022, against three baselines

03

POSITION

Idle units move toward the zones where demand is expected, at least one unit stays in every borough, routes follow real streets.

30 min

between repositioning decisions

04

REPLAY

Real 2022 calls replayed against a fixed layout, same fleet, same calls. Critical calls reached within 8 minutes:

82% → 91%

static layout vs model-guided, 19 July 2022 (modelled)

RUNS ON

IBM LINUXONE

s390x
model trained, replayed and demo served from the VM

33 / 33
automated tests pass on the VM

Modelled
assumed 330-unit fleet; times are not measured

Source: NYC Open Data EMS incident dispatch data; replay of 2022-07-19, 4,694 calls.

PROJECT DISPATCH

A solution to New York EMS response times.

Real calls: Tue, 19 Jul 2022
Response times are modelled

PAUSED
16:08

USER INTERFACE

REPLAY CONTROL

RUN DEMO

PLAY | RESET

Speed: minutes of city time per second

1× | 2× | 4× | 8×

Time 16:08 (129 of 360 min)

VIEW

COMPARE | MODEL | STATIC

Zone shading shows

- MODEL RISK INDEX
- FORECAST CALLS, NEXT HOUR
- WHAT ACTUALLY HAPPENED

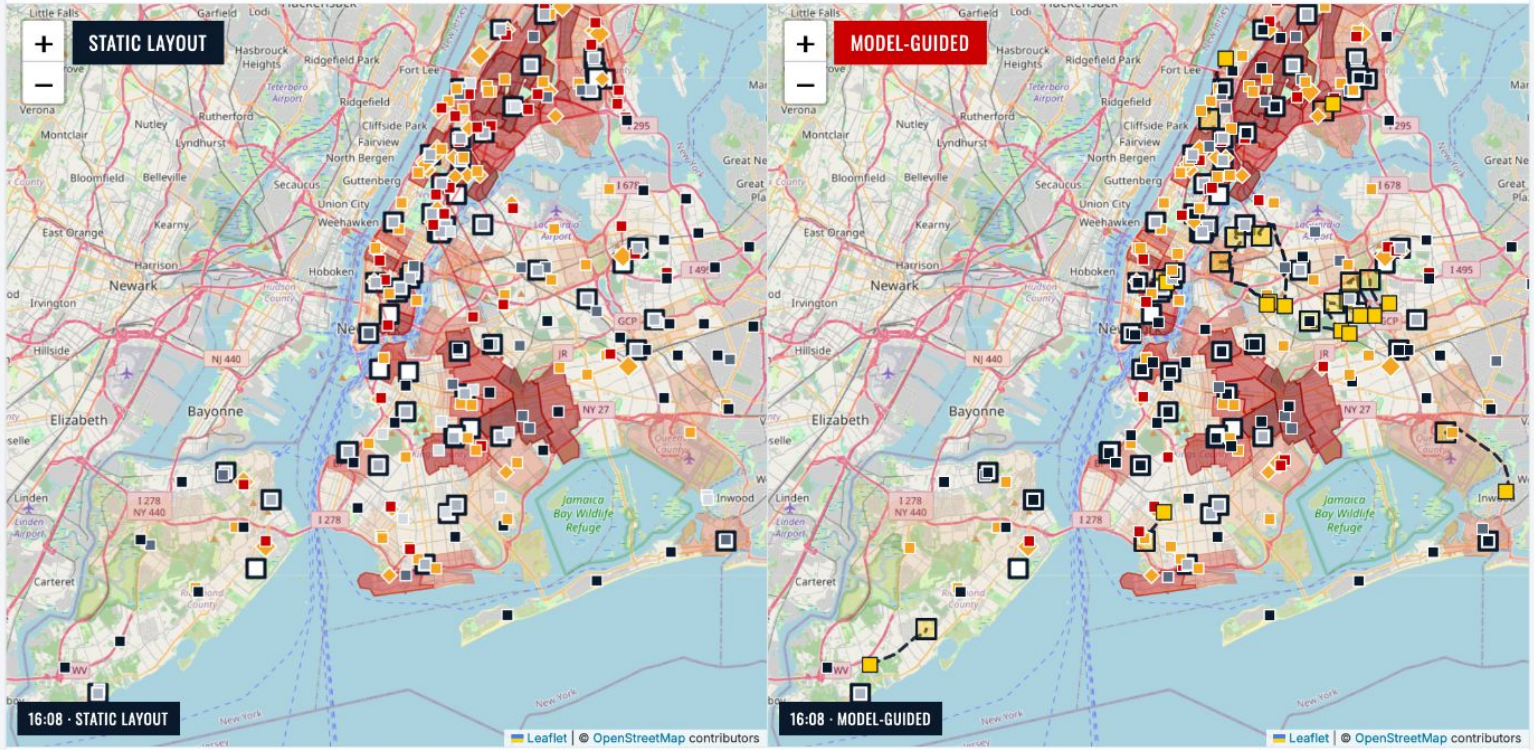
- Zone shading ☑
- Destination lines ☑
- Incoming calls ☑
- Coverage areas ☐

FLEET: 330 UNITS

Available	134
Moving to a new post	15
Responding to a call	28

NYC OPERATIONS MAP

Forecast issued 16:00 · 9 min ago · refreshed every 30 min · shading: model risk index



STATIC LAYOUT

91.8% Critical calls reached within 8 min

4.7 MIN Mean response

MODEL-GUIDED

96.0% Critical calls reached within 8 min

3.9 MIN Mean response

INCOMING CALLS

0	28	502
Waiting	Unit en route	Reached

- 16:08 queens 11419 **critical** — unit en route
- 16:08 bronx 10469 **critical** — unit en route
- 16:08 manhattan 10026 — unit en route
- 16:08 queens 11355 **critical** — unit en route

HIGHEST-RISK ZONES, NEXT HOUR

ZIP ZONE	RISK	STATIC	MODEL
10456 BRO	100	0	2
11212 BRO	99	0	3
10029 MAN	99	0	0
10457 BRO	98	0	0
11207 BRO	98	0	2
10458 BRO	97	0	1
10467 BRO	97	0	1

Units parked in, or driving to, each zone's area. The model moves units toward the zones it expects to be busy.

MODEL DECISIONS

16:00 Unit 0
Sent to post: 11213 to 11225 (brooklyn), 7 min drive. Expected-call coverage +0.41

16:00 Unit 10
Sent to post: 10461 to 10459 (bronx), 19

THE COMPETITIVE ADVANTAGE

LEGACY SOLUTIONS

13 MIN

Mean response time.

- ✕ Static, reactive deployment.

PROJECT DISPATCH

9.65 MIN

Mean response time.

- ✓ 90% of critical calls reached in 8 minutes.
- ✓ Improved coverage in underserved areas.

THE TEAM



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THE FUTURE

BUILDING THE NEXT STANDARD OF URBAN RESPONSE.

Questions & Discussion

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